

Chapter 6

The Spin-Orbit Interaction

Some of this follows §2.3 in Foot [2], but that book provides no explanation of what Thomas precession is.

6.1 Electron Spin

Our investigation of electron orbital angular momentum showed that the simultaneous eigenstates of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$ took non-negative integer values of L and integer values of M_L (which we often wrote simply as M), its projection quantum number. The eigenvalues of these were given by $L(L + 1)$ and M_L , and the commutation relation for the operators was summarized by $\hat{\mathbf{L}} \times \hat{\mathbf{L}} = i\hat{\mathbf{L}}$.

The Stern-Gerlach experiment (among others) showed that the electron has another type of angular momentum, an *intrinsic* angular momentum called *spin*. Unlike orbital angular momentum, electron spin is not characterized by spatial functions such as $Y_{JM}(\boldsymbol{\omega})$, but is instead regarded as a property of the electron itself. The fact that electron spin takes half-integer values also distinguishes it from orbital angular momentum.

To be specific, electron spin states are given by $|S M_S\rangle$ and are simultaneous eigenstates of the (dimensionless) total spin operator squared $\hat{\mathbf{S}}^2$ and the spin projection operator $\hat{S}_z$ with

$$\hat{\mathbf{S}}^2 |S M_S\rangle = S(S + 1) |S M_S\rangle \quad (6.1)$$

and

$$\hat{S}_z |S M_S\rangle = M_S |S M_S\rangle. \quad (6.2)$$

The spin operator itself obeys $\hat{\mathbf{S}} \times \hat{\mathbf{S}} = i\hat{\mathbf{S}}$ and the projection-raising and -lowering operators do the same thing that the orbital versions did, but for spin:

$$\hat{S}_{\pm} |S M_S\rangle = \sqrt{S(S+1) - M_S(M_S \pm 1)} |S, M_S \pm 1\rangle. \quad (6.3)$$

6.1.1 Spin 1/2

For a single electron, we say that it has “spin 1/2” and the projection quantum number M_S can take only two values, $\frac{1}{2}$ and $-\frac{1}{2}$. This is a sufficiently important case that an entire notation has been developed for it, and we often call these states “spin up,”

$$|S = \frac{1}{2}, M_S = +\frac{1}{2}\rangle \equiv |\uparrow\rangle \quad (6.4)$$

and “spin down”

$$|S = \frac{1}{2}, M_S = -\frac{1}{2}\rangle \equiv |\downarrow\rangle. \quad (6.5)$$

Despite the fact that we associate an angular momentum of $\frac{1}{2}\hbar$ with these states, they are really only eigenstates of $\hat{\mathbf{S}}^2$, not $|\hat{\mathbf{S}}|$, and the eigenvalue of $\hat{\mathbf{S}}^2$ is $\frac{3}{4}$. Some people like to take this fairly literally and consider a vector of length $\frac{\sqrt{3}}{2}\hbar$ to represent the spin angular momentum, but I find this a bit confusing and unnecessary. All we really need to do is to work out the algebraic consequences of angular momentum quantum number $J = \frac{1}{2}$:

$$\begin{aligned} \hat{\mathbf{S}}^2 |\uparrow\rangle &= \frac{3}{4} |\uparrow\rangle \\ \hat{\mathbf{S}}^2 |\downarrow\rangle &= \frac{3}{4} |\downarrow\rangle \\ \hat{S}_z |\uparrow\rangle &= \frac{1}{2} |\uparrow\rangle \\ \hat{S}_z |\downarrow\rangle &= -\frac{1}{2} |\downarrow\rangle \\ \hat{S}_{\pm} |S = \frac{1}{2}, M_S\rangle &= \sqrt{\frac{1}{2} \mp M_S} |S = \frac{1}{2}, M_S \pm 1\rangle. \end{aligned}$$

For that last equation, it is perhaps easier to remember if we write a spin 1/2 representation of the operator itself in terms of the two eigenstates as

$$\begin{aligned} \hat{S}_+ &= |\uparrow\rangle\langle\downarrow| \\ \hat{S}_- &= |\downarrow\rangle\langle\uparrow|. \end{aligned}$$

A very useful representation for spin 1/2 systems (or, more generally, for *any* two-level system) is a so-called *spinor* representation, which is a two-dimensional matrix/vector representation with $\hat{S}_z$ eigenstates

$$|\uparrow\rangle \doteq \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (6.6)$$

$$|\downarrow\rangle \doteq \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (6.7)$$

In the spinor representation for spin 1/2, the spin operators are proportional to the Pauli matrices,

$$\begin{aligned} \hat{S}_x &= \frac{1}{2}\sigma_X \\ \hat{S}_y &= \frac{1}{2}\sigma_Y \\ \hat{S}_z &= \frac{1}{2}\sigma_Z \\ \hat{S}_+ &= \sigma_+ \\ \hat{S}_- &= \sigma_- \end{aligned}$$

where

$$\sigma_X \doteq \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_Y \doteq \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_Z \doteq \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (6.8)$$

$$\sigma_+ \doteq \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad \sigma_- \doteq \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \quad (6.9)$$

For an isolated electron spin in a static, uniform magnetic field $\mathbf{B}$, the energies of the $|\uparrow\rangle$ and $|\downarrow\rangle$ states will no longer be degenerate since the electron has a magnetic moment, $\boldsymbol{\mu}_S$, and the interaction Hamiltonian is $H = -\boldsymbol{\mu}_S \cdot \mathbf{B}$. Specifically, the magnetic moment of the electron points along (or “anti-along,” as the case may be) its spin direction,

$$\boldsymbol{\mu}_S = -\mathbf{S} g_s \frac{e\hbar}{2m} \quad (6.10)$$

where $m = m_e$ is the mass of the electron and we will follow the convention in your book that the electron’s “ g factor,” g_s , is positive [2].

The g factor is a dimensionless constant that relates the projection of the spin (M_S) to the value of the projection of the magnetic moment, and is given approximately by

$$g_s \approx 2 + \frac{\alpha}{\pi}$$

where the α/π correction was first derived by Schwinger¹. For our purposes, we will generally use $g_s = 2$.

The second term is the definition of the Bohr magneton,

$$\mu_B \equiv \frac{e\hbar}{2m_e},$$

which corresponds to about 14 GHz/T.

6.1.2 Product States

Much like how we earlier separated the electron's hydrogenic wavefunction into a product of a radial wavefunction, $R(r)$, and an angular wavefunction, $A(\boldsymbol{\omega})$, you may recall that the state kets themselves were also factorable into a radial state and an angular momentum eigenstate,

$$|\psi_{n,L,M_L}\rangle = |R_{n,L}\rangle |L M_L\rangle. \quad (6.11)$$

Spatial wavefunctions are found by taking the inner product with radial position and angular eigenstates $|r\rangle$ and $|\boldsymbol{\omega}\rangle$,

$$\begin{aligned} \langle r | R_{n,L} \rangle &= R_{n,L}(r) \\ \langle \boldsymbol{\omega} | L M_L \rangle &= Y_{L,M_L}(\boldsymbol{\omega}). \end{aligned}$$

Since the electron has spin, this needs to be included in the state vector, and we can start by considering product states of the form

$$|n, L M_L, S M_S\rangle = |R_{n,L}\rangle |L M_L\rangle |S M_S\rangle. \quad (6.12)$$

Note that, despite the similarities between (6.12) and (6.11), this generalization still does not permit a wavefunction to be extracted for the spin degree of freedom. Product states such as this would be energy eigenstates if the spin angular momentum and the orbital angular momentum essentially did not interact with one another.

¹The correction is typically remembered in the form $\frac{g_s-2}{2} = \frac{\alpha}{2\pi}$, hence the “ $\frac{\alpha}{2\pi}$ ” sign on the Schwinger lounge on the 4th floor of PAB.

6.2 Spin-Orbit Coupling

Back when we examined the Bohr model, we estimated the velocity of the electron and found it to be relativistic ($v \approx Z\alpha c/n$), which led to fractional corrections to the energies that scaled like

$$\frac{\Delta E}{E} \approx Z \frac{\alpha^2}{n^2}$$

where α is the fine-structure constant.

Given that we know the velocity of the electron is relativistic, we expect that the Coulomb potential will need to be modified in the electron's frame. In particular, if we have a purely electric field $\mathbf{E}$ in the lab frame, we know that when viewed in a frame moving at velocity $\mathbf{v}$ relative to the lab frame, a magnetic field appears, given by

$$\mathbf{B}' = -\frac{1}{c} \left(\frac{\mathbf{v}}{c} \right) \times \mathbf{E}.$$

Obviously, unless v is substantial compared to c , this relativistic magnetic field is typically ignored.

Since we know the form of $\mathbf{E}$, we can work out this magnetic field,

$$\mathbf{B}' = -\frac{1}{c} \left(\frac{\mathbf{v}}{c} \right) \times \hat{\mathbf{e}}_r \left(\frac{Ze}{4\pi\epsilon_0 r^2} \right)$$

We can simplify the cross product by noting that

$$\begin{aligned} \mathbf{v} \times \hat{\mathbf{e}}_r &= -\hat{\mathbf{e}}_r \times \mathbf{v} \\ &= -\frac{1}{r} \mathbf{r} \times \mathbf{v} \\ &= -\frac{1}{mr} \mathbf{r} \times \mathbf{p} \\ &= -\frac{\hbar}{mr} \mathbf{L} \end{aligned}$$

and we have

$$\mathbf{B}' = \frac{\hbar}{mc^2} \left(\frac{Ze}{4\pi\epsilon_0 r^3} \right) \mathbf{L} \quad (6.13)$$

This classical expression can be quantized using the highly-sophisticated method known as “adding hats to things.”

We can now write the Hamiltonian that will arise due to the interaction of the electron spin and this magnetic field from its orbit, which is known as the spin-orbit interaction:

$$\begin{aligned} H_{\text{SO}} &= -\hat{\boldsymbol{\mu}}_S \cdot \hat{\mathbf{B}}' \\ &= (g_s - 1) \mu_B \frac{\hbar}{mc^2} \left(\frac{Ze}{4\pi\epsilon_0 r^3} \right) \hat{\mathbf{S}} \cdot \hat{\mathbf{L}} \end{aligned} \quad (6.14)$$

$$= (g_s - 1) \mu_B \frac{\hbar^2}{mce} Z\alpha \frac{\hat{\mathbf{S}} \cdot \hat{\mathbf{L}}}{r^3} \quad (6.15)$$

Here, the careful reader will notice that we have replaced the electron spin g factor, g_s , with $(g_s - 1)$ to account for a relativistic effect known as *Thomas precession*. Thomas precession is explained in more detail in §6.3; for the moment, we will take this as given.

Since $\alpha \ll 1$, the relativistic corrections are small compared to the non-relativistic interactions and we can treat the spin-orbit interaction as a perturbation on the atomic Hamiltonian and use perturbation theory to calculate its magnitude. Specifically, we will be interested in the expectation value

$$\begin{aligned} E_{\text{SO}} &= \langle H_{\text{SO}} \rangle \\ &= (g_s - 1) \mu_B \frac{\hbar^2}{mce} Z\alpha \left\langle \frac{\hat{\mathbf{S}} \cdot \hat{\mathbf{L}}}{r^3} \right\rangle \end{aligned} \quad (6.16)$$

for which we need to think about the states in which we are to calculate this. As before, the capital-letter operators $\hat{\mathbf{S}}$ and $\hat{\mathbf{L}}$ are dimensionless and correspond to the true, dimensionful angular momentum operators divided by $\hbar$.

6.2.1 Total Angular Momentum Eigenstates

Since the spin-orbit Hamiltonian, Eq. (6.15), takes different expectation values for states with different expectation values for $\hat{\mathbf{S}} \cdot \hat{\mathbf{L}}$, we expect the energy eigenstates to be characterized by the value of $\hat{\mathbf{S}} \cdot \hat{\mathbf{L}}$, which may mean they are no longer characterized by individual projection quantum numbers M_S and M_L . As a consequence, product states of the form given in Eq. (6.12) may not be energy eigenstates, and we may need to form linear combinations of

these to find the eigenstates of the system (the procedure for doing this will be covered in Chapter 9).

More broadly, the isotropy of space tells us that the total angular momentum of an atom in free space is conserved, and we might therefore expect J , the total angular momentum quantum number, to be a good quantum number. We can construct a (dimensionless) total angular momentum *operator* by taking the vector sum of the orbital and spin angular momentum operators,

$$\mathbf{J} = \mathbf{L} + \mathbf{S}, \quad (6.17)$$

(where I no longer care to keep adding hats to denote operators, which I believe the reader can by this point keep track of on their own). Since the atomic Hamiltonian (even with the spin-orbit interaction) is rotationally invariant, we see that it commutes with $\mathbf{J}^2$ and J_z , and the angular eigenstates are therefore no longer strictly associated with values of M_L or M_S . Instead, they are eigenstates $|LS, JM_J\rangle$ of $\mathbf{L}^2$, $\mathbf{S}^2$, $\mathbf{J}^2$ and J_z (with eigenvalues $L(L+1)$, $S(S+1) = \frac{3}{4}$, $J(J+1)$ and M_J).

When we get to multi-electron atoms, we will see that L and S are no longer characteristic quantum numbers for multi-electron atomic states, though they are still good for single-electron atoms. In the spirit of perturbation theory, we generally still associate them with the multi-electron atomic eigenstates, essentially by taking a correspondence with what the true states would be in the limit as $H_{\text{SO}} \rightarrow 0$. The individual projection quantum numbers, M_S and M_L , however, can be highly heterogeneous for the atomic eigenstates (even for single-electron atoms) and these are no longer descriptive of the actual eigenstates². The atomic eigenstates will therefore be written with these labels,

$$|n, LS, JM_J\rangle = |R_{n,L}\rangle |LS, JM_J\rangle. \quad (6.18)$$

It is no longer necessarily possible to factor this further into a spin state and an orbital state, as these can now be *entangled*.

With these basis states, we can calculate the expectation value of $\mathbf{S} \cdot \mathbf{L}$ in terms of the good quantum number J and the (possibly only approximately-) good quantum numbers S and L by using what is effectively just the law of

² M_L and M_S can become approximately good again in very strong external magnetic fields.

cosines:

$$\begin{aligned}
\mathbf{S} \cdot \mathbf{L} &= (\mathbf{J} - \mathbf{L}) \cdot (\mathbf{J} - \mathbf{S}) \\
&= \mathbf{J}^2 - \mathbf{J} \cdot \mathbf{S} - \mathbf{L} \cdot \mathbf{J} + \mathbf{L} \cdot \mathbf{S} \\
&= \mathbf{J}^2 - (\mathbf{L} + \mathbf{S}) \cdot \mathbf{S} - \mathbf{L} \cdot (\mathbf{L} + \mathbf{S}) + \mathbf{L} \cdot \mathbf{S} \\
&= \mathbf{J}^2 - \mathbf{L} \cdot \mathbf{S} - \mathbf{S}^2 - \mathbf{L}^2 - \mathbf{L} \cdot \mathbf{S} + \mathbf{L} \cdot \mathbf{S} \\
&= \mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2 - \mathbf{L} \cdot \mathbf{S} \\
&= \frac{1}{2}(\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2).
\end{aligned}$$

The expectation value can now be taken for the atomic eigenstates (6.18),

$$\begin{aligned}
\langle \mathbf{S} \cdot \mathbf{L} \rangle &= \langle n, LS, JM_J | \mathbf{S} \cdot \mathbf{L} | n, LS, JM_J \rangle \\
&= \frac{1}{2} \langle n, LS, JM_J | (\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2) | n, LS, JM_J \rangle \\
&= \frac{1}{2} (J(J+1) - L(L+1) - S(S+1)).
\end{aligned}$$

Since the expectation of $\frac{1}{r^3}$ only depends on $|R_{n,L}\rangle$ and the expectation value of $\mathbf{S} \cdot \mathbf{L}$ only depends on $|LS, JM_J\rangle$, the angular and radial parts of the state act separately,

$$\begin{aligned}
\left\langle \frac{\mathbf{S} \cdot \mathbf{L}}{r^3} \right\rangle &= \langle n, LS, JM_J | \left(\frac{\mathbf{S} \cdot \mathbf{L}}{r^3} \right) | n, LS, JM_J \rangle \\
&= \langle R_{n,L} | \langle LS, JM_J | \left(\frac{\mathbf{S} \cdot \mathbf{L}}{r^3} \right) | LS, JM_J \rangle | R_{n,L} \rangle \\
&= \langle R_{n,L} | \frac{1}{r^3} | R_{n,L} \rangle \langle LS, JM_J | \mathbf{S} \cdot \mathbf{L} | LS, JM_J \rangle \\
&= \frac{1}{2} (J(J+1) - L(L+1) - S(S+1)) \langle R_{n,L} | \frac{1}{r^3} | R_{n,L} \rangle.
\end{aligned}$$

The radial integral for the expectation value can be computed for individual radial wavefunctions (assuming $L \neq 0$ – we expect no spin-orbit interaction in states with no orbital angular momentum), but there is also a general formula that can be computed, and I will simply state that, for hydrogenic radial states,

$$\begin{aligned}
\langle R_{n,L} | \frac{1}{r^3} | R_{n,L} \rangle &= \int_0^\infty dr r^2 \frac{1}{r^3} |R_{n,L}(r)|^2 \\
&= \frac{1}{a_0^3 L(L + \frac{1}{2})(L+1)} \frac{Z^3}{n^3}.
\end{aligned} \tag{6.19}$$

From this, the spin-orbit splitting can be calculated in hydrogen, and is given by

$$E_{\text{SO}}(n, L, S, J) = (g_s - 1) \frac{Z^4 \alpha^2 h c R_\infty}{2n^3} \times \frac{J(J+1) - L(L+1) - S(S+1)}{L(L + \frac{1}{2})(L+1)}. \quad (6.20)$$

For the $2p$ states of hydrogen, this gives $E_{\text{SO}} = 10.975$ GHz. The measured value is $E_{\text{SO}} = 10.969$ GHz, so this is good to 5×10^{-4} . Effects that we have not considered include higher-order relativistic corrections and energy shifts from the vacuum fluctuations of the electromagnetic field (the most famous of which is known as the *Lamb shift*).

6.2.2 Fine Structure States

Due to the spin-orbit interaction, energy eigenstates with the same values of the quantum numbers L and S but different values for the quantum number J can have different energy. This splitting is historically called “fine structure,” and the prototypical case is the splitting between the $J = \frac{1}{2}$ and $J = \frac{3}{2}$ manifolds of the $2p$ states in hydrogen.

In the more general case, if we are given S and L , we can list the allowed values of J by starting from $J_{\text{max}} \equiv L + S$ and stepping down J in steps of 1 until we get to $|L - S|$. So for a 2P_J manifold, we have $L = 1$ (from the “P”) and $S = \frac{1}{2}$ (from the $2S + 1 = 2$ leading superscript), and we see that the two allowed values (which we’ll call J_{max} and J') are

$$\begin{aligned} J_{\text{max}} &= L + S = 1 + \frac{1}{2} = \frac{3}{2} \\ J' &\equiv J_{\text{max}} - 1 = \frac{1}{2}. \end{aligned} \quad (6.21)$$

The process in this case terminates at $J' = \frac{1}{2}$ since this is $|L - S|$.

As another example, a manifold described by the term symbol $^4F_J^o$ would have 4 separate energy eigenstates, with J taking values from $\frac{9}{2}$ down to $\frac{3}{2}$ (as always, in steps of $\Delta J = 1$).

6.3 Thomas Precession

The phenomenon of “Thomas precession” was invoked (without explanation) to account for the fact that the g factor of the electron spin needed to be

modified from g_s to $(g_s - 1)$ when calculating the fine-structure splitting. The splitting itself came about from spin precession in the magnetic field that appeared in the electron's rest frame due to its relativistic motion in the electric field of the nuclear charge. This section is meant to provide more explanation of the origin of the Thomas precession, but in the simplest possible terms³.

Thomas precession is the name given to a relativistic frame rotation that accompanies two, non-co-linear Lorentz boosts. To illustrate the frame rotation, we consider a train car that is whizzing counter-clockwise around a closed loop of track at a relativistic speed. Though the track may appear circular from far away, we can consider that it is in fact a regular polygon with N edges, fixed in the xy plane, for some large integer $N \gg 1$. Each edge of the polygon is a straight section of track and each corner involves turning the train car counter-clockwise to align it with the next section of straight track.

From the Earth-fixed frame (we will call this the “lab frame”), the train turns through an angle (see Fig. 6.1)

$$\theta \approx \frac{W}{L} \approx \frac{2\pi}{N} \quad (6.22)$$

at each of the N corners such that after a lab-frame time interval T , the train has orbited all the way around the track. The orientation of the train in the xy plane has therefore spun at an approximately constant rate of

$$\boldsymbol{\omega}_0 \equiv \frac{2\pi}{T} \hat{\mathbf{e}}_z. \quad (6.23)$$

For observers on the train, however, things may not be the same. For example, as the engineer who is driving the train approaches corner number j , she can look ahead to corner $j + 1$ to determine how much the train car needs to spin at the corner so that it lines up with the next section of straight track. Figure 6.1 shows that as the train approaches the corner, the engineer will see that corner $j + 1$ appears a perpendicular distance W to the left of the train's current path, while the distance along the train's path, which is L in the lab frame, is Lorentz contracted to $L' \equiv L/\gamma$ in the train's frame (where $\gamma = 1/\sqrt{1 - v^2/c^2}$).

³This follows an elegant explanation attributed to Ed Purcell [4].

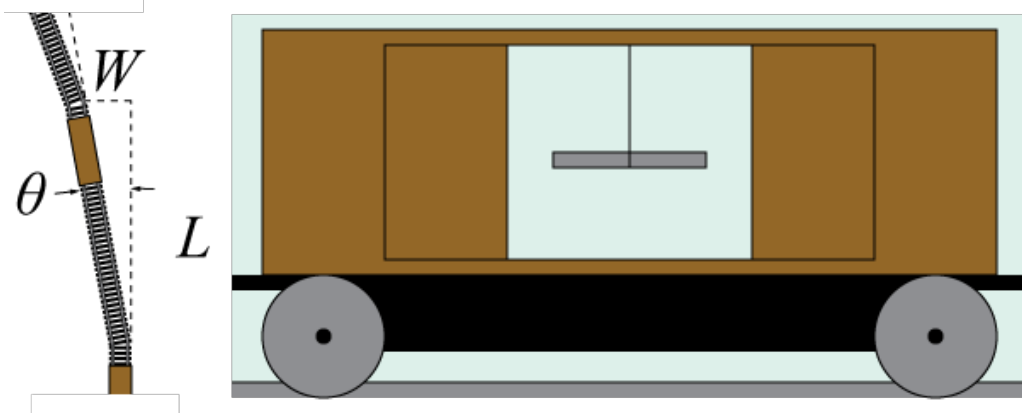


Figure 6.1: A train car traveling around a closed loop of track in the shape of a regular N -gon. At the end of each straight segment, the velocity vector's direction changes by θ in the lab frame. A passenger in the train car can measure the rotation by observing the apparent rotation of a bar hung freely from the roof of the car.

So the engineer turns the car through angle θ' at each corner, where

$$\theta' \approx \frac{W}{L/\gamma} \approx \gamma \theta.$$

This is not just an optical illusion. If one of the passengers has decided to measure the train car's orientation with respect to the lab frame by hanging a rod from the ceiling of the train car by its center of mass, he will also conclude that the train turns through angle $\theta' = \gamma \theta$ at each corner. Since the relationship between θ and θ' does not depend on N , we can now take $N \rightarrow \infty$ to converge to a circular trajectory.

After a complete lap, the people on the train will have experienced a rotation not by 2π , but rather by $\gamma 2\pi$. The “extra” rotation means that an observer in the lab frame who examines the angle of the hanging rod will see it rotate in the $-\hat{\mathbf{e}}_z$ direction. At each turn, the bar will rotate in the lab frame by angle $\gamma\theta - \theta$ in the clockwise direction. In the infinite N limit, from the lab frame, the bar appears to spin at a constant rate of

$$\begin{aligned} \boldsymbol{\omega}_T &= \frac{\gamma 2\pi - 2\pi}{T} (-\hat{\mathbf{e}}_z) \\ &= -(\gamma - 1) \frac{2\pi}{T} \hat{\mathbf{e}}_z. \end{aligned}$$

As a fraction of the orbital speed, ω_0 , this is

$$\begin{aligned}\frac{\omega_T}{\omega_0} &= \gamma - 1 \\ &= \frac{1}{\sqrt{1 - \beta^2}} - 1 \\ &\approx \frac{1}{2} \frac{v^2}{c^2}.\end{aligned}\tag{6.24}$$

The rotation,

$$\boldsymbol{\omega}_T \approx -\frac{1}{2} \frac{v^2}{c^2} \boldsymbol{\omega}_0,\tag{6.25}$$

of the reference frame that is orbiting the origin at a relativistic speed is known as Thomas precession.

6.3.1 Application to electron spin

For the electron in a classical, circular Bohr orbit confined to the xy plane with (dimensionless) orbital angular momentum $\mathbf{L} = L \hat{\mathbf{e}}_z$, we showed that special relativity dictates that in the electron's frame, a magnetic field appears along $\mathbf{L}$ that is given by Eq. (6.13).

We showed that the Thomas precession is antiparallel to the orbital angular momentum, but what about the orientation of the spin precession? If we consider for a moment a classical spin and that the electron spin points along $\hat{\mathbf{e}}_x$ at some instant t_0 in the electron's rest frame, the rate at which this spin precesses (sometimes called *Larmor precession*) around the z axis is the frequency shift we are trying to calculate, as this is the frequency required to flip the spin between $\pm \hat{\mathbf{e}}_z$. Classically, the magnetic field exerts a torque on the magnetic dipole given by

$$\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}.\tag{6.26}$$

Since the electron's spin points the opposite direction from its magnetic moment, we have $\boldsymbol{\mu}_s \parallel -\hat{\mathbf{e}}_x$. The instantaneous direction of the torque, then, is given by

$$\begin{aligned}\frac{d\boldsymbol{\tau}}{|d\boldsymbol{\tau}|} &= -\frac{\mathbf{S}}{|\mathbf{S}|} \times \frac{d\mathbf{L}}{|d\mathbf{L}|} \\ &= -\hat{\mathbf{e}}_x \times \hat{\mathbf{e}}_z \\ &= \hat{\mathbf{e}}_y\end{aligned}\tag{6.27}$$

and we see that the spin precesses in the electron's frame about $\mathbf{B}'$, which is parallel to the orbital angular momentum and therefore anti-parallel to the Thomas precession. The rate of this Larmor precession in the electron's rest frame is

$$\boldsymbol{\omega}'_{\text{L}} = \mu_s \mathbf{B}'. \quad (6.28)$$

To get the precession rate of the spin as seen by an observer in the lab frame ($\boldsymbol{\omega}_{\text{L}}$), we need to add the Thomas precession,

$$\begin{aligned} \boldsymbol{\omega}_{\text{L}} &= \boldsymbol{\omega}'_{\text{L}} + \boldsymbol{\omega}_{\text{T}} \\ &\approx \boldsymbol{\omega}'_{\text{L}} - \frac{1}{2} \frac{v^2}{c^2} \boldsymbol{\omega}_0 \end{aligned}$$

where $\boldsymbol{\omega}_0$ is the angular velocity of the orbital motion. Since the spin precession rate ω_{L} (times $\hbar$) is equivalent to the energy difference between spin up and spin down, we can write the effective Hamiltonian as

$$\begin{aligned} H_{\text{SO}} &= \mathbf{S} \cdot (\hbar \boldsymbol{\omega}_{\text{L}}) \\ &= \hbar \mathbf{S} \cdot \boldsymbol{\omega}'_{\text{L}} - \frac{\hbar v^2}{2 c^2} \mathbf{S} \cdot \boldsymbol{\omega}_0 \\ &= -(\boldsymbol{\mu}_s \cdot \mathbf{B}') - \frac{\hbar v^2}{2 c^2} \mathbf{S} \cdot \boldsymbol{\omega}_0 \end{aligned}$$

We can recognize the first term as the spin-orbit Hamiltonian without the Thomas precession correction, and use the Bohr model to write $\boldsymbol{\omega}_0 = \frac{\mathbf{L}}{mr^2}$, which gives us

$$H_{\text{SO}} = g_s \mu_{\text{B}} \frac{\hbar^2}{m c e} Z \alpha \frac{\mathbf{S} \cdot \mathbf{L}}{r^3} - \frac{\hbar v^2}{2 c^2} \frac{\hbar}{m r^2} \mathbf{S} \cdot \mathbf{L}.$$

Next, we can use an expression for v^2 from the Bohr model,

$$v^2 = Z \alpha \frac{\hbar c}{m r}$$

to write the Hamiltonian as

$$\begin{aligned} H_{\text{SO}} &= g_s \mu_{\text{B}} \frac{\hbar^2}{m c e} Z \alpha \frac{\mathbf{S} \cdot \mathbf{L}}{r^3} - \frac{\hbar}{2 c^2} \frac{1}{m r^2} \left(Z \alpha \frac{\hbar c}{m r} \right) \mathbf{S} \cdot \mathbf{L} \\ &= g_s \mu_{\text{B}} \frac{\hbar^2}{m c e} Z \alpha \frac{\mathbf{S} \cdot \mathbf{L}}{r^3} - \mu_{\text{B}} \frac{\hbar^2}{m c e} Z \alpha \frac{\mathbf{S} \cdot \mathbf{L}}{r^3} \\ &= (g_s - 1) \mu_{\text{B}} \frac{\hbar^2}{m c e} Z \alpha \frac{\mathbf{S} \cdot \mathbf{L}}{r^3} \end{aligned}$$

which is how we obtained Eq. (6.15).

The appeal to the Bohr model in this derivation came strictly from our use of the train model for calculating the relativistic frame rotation; a fully relativistic version of the quantum calculation gives the same result when truncated to this order (with considerably more computational complexity), and a more-careful classical treatment would yield the correction to the Bohr model to higher order in v/c . In contrast, what we've done is to essentially fake our way to an effective Hamiltonian (H_{SO}) that can be used to account for some relativistic effects phenomenologically.

Last, note that Thomas precession is not very electromagnetic in its origin. It is certainly *consistent* with electromagnetism, but its appearance as a modification of the electron's spin g factor is a little misleading. The electron's g factor is still g_s , even with this relativistic motion (at least at first order), but a correction to the spin-orbit shift arises that can be added phenomenologically by subtracting 1 from the g factor (but essentially *only* for calculating the spin-orbit interaction).